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Cost-Effective Chemical Upgradation of Plastic Pyrolysis Oil for Engine Applications: Performance Evaluation as a Diesel Blend

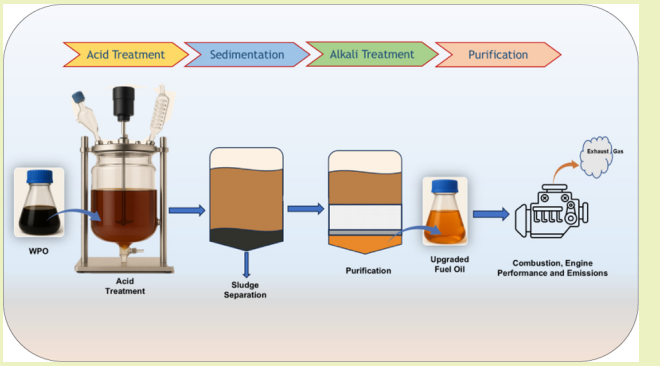
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Cite This: *ACS Sustain. Resour. Manag.* 2026, 3, 2702−2713

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ABSTRACT: This study evaluates upgraded waste plastic pyrolysis oil (UWPO) as a sustainable alternative fuel to address the dual challenges of increasing energy demand and plastic waste accumulation. The novelty of this work lies in the development of an energy-efficient and cost-effective chemical upgrading strategy that enhances the fuel quality of waste plastic pyrolysis oil without relying on conventional high-pressure hydrogenation or energy-intensive refining processes, thereby improving its suitability for compression ignition engines. The upgraded oil was comprehensively characterized for key physicochemical properties, including density, gross calorific value, flash point (10 °C), fire point (14 °C), pour point (−4 °C), and viscosity (1.08 cSt), demonstrating improved fuel handling and storage characteristics. Engine performance, combustion, and emission analyses were conducted using 10% (UWPO10) and 20% (UWPO20) UWPO blends with commercial diesel. Comparative evaluation showed that commercial diesel exhibited the highest brake thermal efficiency (33.86%), whereas UWPO20 achieved 27.80%, indicating a moderate reduction in efficiency at higher blend ratios. Similarly, brake specific fuel consumption increased from 6.353×10^{-5} kg kW^{-1} s^{-1} for diesel to 7.804×10^{-5} kg kW^{-1} s^{-1} for UWPO20. However, UWPO10 exhibited the highest mechanical efficiency (92.87%), surpassing both commercial diesel (87.47%) and UWPO20 (88.59%), while both blends showed reduced air−fuel ratios, indicating improved combustion characteristics. To evaluate industrial feasibility, a techno-economic analysis of a 10 TPD waste plastic pyrolysis and oil upgrading facility was performed, demonstrating the economic viability of the proposed process with favorable production economics compared with conventional fossil-derived fuels. Overall, the study establishes chemically upgraded waste plastic pyrolysis oil as a promising diesel blending component and provides an integrated technical and economic framework for sustainable plastic waste valorization, supporting circular economy and low-carbon energy initiatives.



KEYWORDS: *plastic pyrolysis oil, chemical upgradation, engine performance analysis, combustion analysis, emission analysis*

1. INTRODUCTION

The escalating global challenge of plastic waste management, amid the global shift toward sustainable energy sources, has inspired research into innovative solutions. The rising demand is primarily a result of growing recognition of the need for alternative, sustainable energy sources. A promising solution to these issues is waste plastic pyrolysis oil (WPO), a fuel produced by the thermal decomposition of plastic waste. According to a recent market analysis, the global waste plastic pyrolysis oil market (valued at USD $641.1 M in 2021) is projected to experience steady growth (at a CAGR of 5.3% until 2032) driven by increasing interest in sustainable waste-to-fuel technologies and circular resource management approaches.[1,2] Moreover, global events such as the COVID-19 pandemic and the Russia−Ukraine conflict have once again highlighted the world's reliance on conventional fossil fuels and the instability in their prices. According to a report by Global Market Insights, the application of WPO is divided into generator fuel, boilers for heat, chemical extraction, and other minor categories. However, in order to reduce dependency on fossil fuels, the use of WPO to substitute or supplement automotive fuels is necessary.

Currently, WPO faces numerous challenges such as non-uniform composition due to unstandardized feed (varying compositions of polymers in waste), low pH values corresponding to high acidity, which can lead to corrosion, high unsaturation

Received: March 16, 2026
Revised: July 2, 2026
Accepted: July 2, 2026
Published: August 11, 2026



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that causes sticky deposits, increasing the cost of handling, wax formations in oil at lower temperatures, causing storage problems in cold-regions, non-homogenous nature, leading to layering or partial separation, the presence of suspended carbon, reducing the calorific value of oil, along with causing depositions in boilers, unpleasant odor, lower flash point, lower cetane index, and the high viscosity of plastic pyrolysis oils, making it difficult to be transported through pipelines, increasing the cost of distribution and selling.[3] These listed reasons make it difficult for unprocessed and non-purified WPO to be directly used in automotive applications. Researchers have explored several techniques such as catalytic cracking, distillation, hydrogenation, steam reforming, hydrodeoxygenation, and electrochemical hydrogenation to upgrade the WPO quality. Conventional upgrading methods like distillation and hydrothermal cracking can be energy-intensive and economically challenging. Table 1 provides a summary of different WPO upgradation processes to obtain upgraded oil.

Dobó et al. carried out atmospheric distillation of pyrolysis oil obtained from various polymer feeds and classified the resulting oil into gasoline and diesel fractions, with PS reported to yield 70.5% gasoline.[4] Arjharn et al. obtained 60% diesel oil through vacuum distillation of mixed plastic waste-derived pyrolysis oil at 282 °C.[5] Mangesh et al. conducted hydrogenation of PP-derived pyrolysis oil using a modified Ni-Mo/Lap catalyst to achieve 95% composition of oil in the diesel range.[7] Faisal et al., in their work, compared the impact of distillation and hydrotreatment on pyrolysis oils of both waste and virgin plastic mixtures on the grounds of automobile fuel specifications.[15] The distillation and hydro-processing are conventionally studied for WPO upgradation; however, issues such as high energy requirement, high pressure, high temperature, hydrogen source, and large investment make the process cost too high, making them commercially unviable. An alternative approach wherein Catizane et al. reviewed that enhancing the quality of pyrolysis oil made from plastic and biomass through electrochemical hydrogenation and compared to conventional methods like thermal cracking and electrochemical hydrogenation may be carried out without the need for an additional H_2 supply and under mild conditions (80 °C and 1 atm) in an electrolysis cell separated with proton exchange membranes.[16] Challenges with this approach are its applicability for feedstock with varying composition, scalability, and membrane fouling with carbon deposition.

Furthermore, the use of pyrolysis oil has been explored in engines, and their performance has been reported. Kalargaris et al. studied on the DI diesel engine characteristics (combustion, performance, and emissions) fueled by plastic pyrolysis oil and reported that blending of pyrolysis oil with diesel and additives resulting in good engine performance with a blend of 60–70% PPO operating at 80–90% engine load seems to have the most promising potential to achieve the best engine performance and less emissions than alone running with pyrolysis oil resulting in incomplete combustion, more of CO formation and reduced engine performance.[17] In a similar approach, Pumpuang et al. investigated on the influence of HDPE- and PP-derived pyrolysis oils, exhibiting the similar fuel properties within the fuel range of diesel, and reported that HDPE exhibits similar combustion characteristics; in contrast, PP displayed significant combustion retardation owing to its lower oxygen content, higher viscosity, and lower density.[18] HDPE yields higher NO_X emissions than diesel at low and medium engine loads but lower emissions at high loads, also they reported that blending this fuel enhances lubricity, engine performance, and combustion characteristics.

The present research aims to overcome the above challenges by investigating a cost-comprehensive approach of chemical treatment to WPO. This tailored process aims to enhance its fuel quality and overcome the limitations of conventional methods, thereby aiming to produce a fuel with improved characteristics suitable for blending with conventional diesel. The upgraded WPO (UWPO) is then

Table 1. State of Art for Upgradation Techniques for Crude Plastic Pyrolysis Oil[a]

pyrolysis oil of	upgradation process	catalyst	operating conditions	outcome comments	ref
HDPE	atmospheric distillation	NR	20–200 °C	37.3%—G; 30.5%—CD	4
LDPE				40.8%—G; 15.2%—CD	
PP				42.1%—G; 24.8%—CD	
PS				70.5%-G; 1.1%-CD	
MPW	vacuum distillation	NR	0.05 bar; 282 °C	60%—CD; 30%—naphtha	5
PE	distillation	NR	180 °C	NR	6
MPW			150–180 °C	NR	
PP	hydrogenation	Ni/ZSM-5	70 bar H_2; 350 °C	75%—CD	7
PP	hydrogenation	Ni-Mo/Lap	70 bar H_2; 350 °C	95%—CD	8
LDPE	hydrogenation	Pd/HZSM-5	20 bar H_2; 250–350 °C	55%—G; 45%—CD	9
PC + PE + PP	catalytic pyrolysis + deoxygenation	HZSM-5	500–700 °C	oxygenates decreased to 2.9%; aromatics increased to 97.1%	10
PC	hydrodeoxygenation	Ni/HZSM-5	40 bar H_2; 190 °C	99.3%—cycloalkanes	11
PC	catalytic pyrolysis + hydro-reforming	Ni/h-beta	20 bar H_2; 310 °C	52–54%—G	12
MPW	hydrogenation	Ni/SiO_2–Al_2O_3	52 bar H_2; 350 °C	85% hydrocarbon; 15% gases	13
MPW	distillation + hydrotreatment	Pd catalyst	210–350 °C	G cut (210 °C); D cut (350 °C)	14

[a]G, gasoline; CD, commercial diesel; MPW, mixed plastic waste; PP, polypropylene; LDPE, low-density polyethylene; PS, polystyrene; PE, polyethylene; PC, polycarbonate.

https://doi.org/10.1021/acssusresmgt.6c00163
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characterized in terms of its key fuel properties and engine performance parameters such as brake thermal efficiency, brake specific fuel consumption, and mechanical efficiency, as well as investigating the combustion characteristics and emission profiles of the blended fuels.

The effectiveness of the chemical upgrading process depends strongly on the composition of the plastic feedstock used for pyrolysis. Pyrolysis oils derived primarily from polyolefin plastics, such as polyethylene (PE) and polypropylene (PP), generally contain higher concentrations of olefinic and unsaturated hydrocarbons, which are more susceptible to protonation, oligomerization, and condensation reactions during acid treatment. Therefore, such feedstocks are expected to respond favorably to the upgrading process. In contrast, plastics containing heteroatoms, such as polyethylene terephthalate (PET), polyurethane (PU), or nitrogen-containing polymers, may generate pyrolysis oils containing chlorine-, oxygen-, or nitrogen-containing compounds. These species may influence upgrading efficiency, increase corrosive behavior, and require additional pretreatment or purification steps prior to chemical upgrading. Consequently, feedstock variability is an important consideration for large-scale implementation and optimization of the proposed upgrading strategy.

The uniqueness of this study lies in its focus on chemically upgrading waste plastic pyrolysis oil (WPO) through cost-effective treatments to improve its suitability as a diesel blend fuel. Unlike most previous works that examine raw or minimally treated pyrolysis oils, this research systematically enhances fuel properties such as density, calorific value, viscosity, and ignition parameters, enabling more reliable engine operation. Furthermore, the study explicitly integrates detailed performance, combustion, and emission analyses of upgraded WPO (UWPO)-diesel blends under realistic engine conditions, providing comparative insights with commercial diesel. By quantifying changes in brake thermal efficiency, brake specific fuel consumption, mechanical efficiency, and emission behavior, the research delivers a comprehensive understanding of both the advantages and trade-offs of using upgraded WPO (UWPO). This combined approach of chemical upgradation and practical engine testing establishes a novel pathway toward valorizing plastic waste into sustainable fuels, bridging a critical gap between laboratory-scale oil upgrading and its potential real-world application.

2. MATERIAL AND METHODS

2.1. Materials

The crude pyrolysis oil used as feedstock in this study was prepared using the pyrolysis technique reported in our earlier studies (Munagala et al.,[19] Razak et al.,[20] and Munagala et al.[21]). The feed used for pyrolysis is a mixed plastic waste comprising majorly HDPE, PP, LDPE, and PET. The chemical reagents employed for the upgradation of waste pyrolysis oil (WPO) included sulfuric acid (SA, 98% purity), hydrochloric acid (HCl, 36%), and acetic acid (AA, 99%), all procured from Finar (Actylis Lab). In addition, methane sulfonic acid (MSA, 98%) and triethylamine (TEA, 99%) were obtained from SRL.

2.2. WPO Upgradation Process

The upgradation of WPO was carried out utilizing the chemical treatment process, as shown in Figure 1. This process is conducted at 55 °C, with variable concentrations of chemical treating agents such as SA, HCl, AA, and MSA. The treatment was continued for 1 h under continuous stirring conditions. Post the reactions, the oil was left to cool to allow the separation of the heavy sludge layer. The lighter oil was taken out through decantation, was treated with TEA for neutralization, and was passed through a CaO/Na_2SO_4 bed for the removal of any residual acidic species and moisture.

Sulfuric acid (SA) concentrations ranging from 1–10 wt % relative to WPO were investigated to evaluate the effect of acid loading on fuel upgrading behavior. Similar concentration ranges were examined for the other acids investigated in the study. Among the various acid-treated oils investigated (SA, HCl, AA, and MSA), sulfuric-acid-treated WPO exhibited the most favorable physicochemical properties and was therefore selected for engine performance, combustion, and emission studies. The term 'UWPO' used throughout the manuscript refers specifically to this sulfuric-acid-treated upgraded oil.

2.3. Characterization of Oil

The WPO and UWPO samples were characterized through ultimate analysis for estimation of hydrocarbon content using a Vario Micro Cube Elemental analyzer. The API, density, and specific gravity of WPO and UWPO samples were estimated through ASTM D1298, ASTM D1217, and ASTM D287 standard methods. The dynamic viscosity measured using a Borosil Mansingh Survismeter was used to calculate the kinematic viscosity. The oil's chemical composition was analyzed using an Agilent 6890 GC coupled with a 5973 N mass-selective detector and an HP-5MS capillary column (30 m length, 250 μm inner diameter, 0.25 μm film thickness). The column oven's temperature was programmed from 50 °C (2 min hold) to 280 °C at

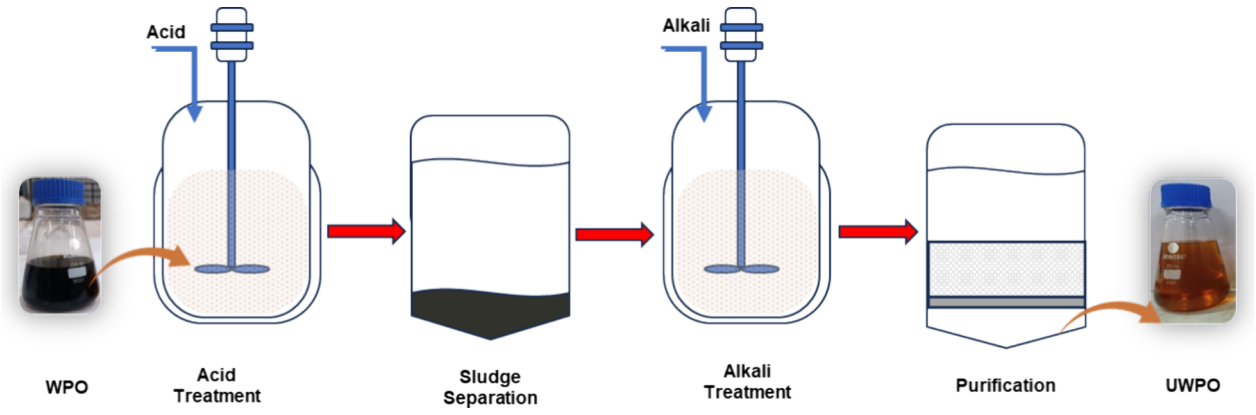


Figure 1. Process flow diagram for the oil upgradation.

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10 °C/min (5 min hold). Helium was the carrier gas at a constant flow of 1 mL/min. The inlet and GC-MS interface were maintained at 250 and 280 °C, respectively, with a detector temperature of 270 °C. The electron impact ionization source and quadrupole analyzer were set at 230 and 150 °C, respectively.[22] Mass spectrometry covered m/z 29–600, and fragment identification was done using the Wiley Registry, 10th Edition, and NIST 2012, following ASTM D6730 standards. The flash and fire points were estimated following ASTM D 93 and GCV by a bomb calorimeter following ASTM D240 standards.

2.4. Engine Performance Tests

The engine performance, combustion, and emission analyses of waste plastic pyrolysis oil (WPO) obtained through waste packaging materials have been carried out in a single-cylinder four-stroke DI diesel engine as per our previous studies.[23,24] The details of the engine specifications are given in Table 2. Engine experiments were carried out with blending of UWPO with commercial diesel (CD) with two blending ratios, viz. 10 and 20% (v/v). The samples were coded as CD100 for pure commercial diesel, UWPO10 for (UWPO 10% + CD 90%) and UWPO20 for (UWPO 20% + CD 80%). The engine performance and combustion analyses were performed using all three fuels at various loading conditions between 0 and 100%. The emissions of exhaust gases released from the engine were analyzed with the help of an AVL 444-N Digas Analyzer.

3. RESULTS AND DISCUSSION

3.1. Treatment Effect on Yield

The treatment of WPO with different chemical agents, such as SA, MSA, AA, and HCl, resulted in the separation of heavy sludge and suspended carbon. The acid reacts with alkenes or dienes and aromatics present in WPO, leading to polymerization and forming heavy tar-like black sludge. This sludge settles at the bottom, while the light fraction of oil forms the top layer. Further alkaline treatment of the top liquid layer neutralizes the residual acidity in the oil, leading to the formation of water-soluble salts, which can be easily separated from the oil, also breaking down larger molecules into smaller ones, increasing the oil's fluidity and reducing its density and viscosity. The effect of acid loading in the oil is shown in Figure 2. The treatment with 1% SA exhibited higher yields than the other acidified oils. It is observed that as the loading of the acid increases, the yields begin to reduce due to the degradation or polymerization of the oil, which is caused by the side reaction of oil with acid (esterification). The treatment of SA, AA, and MSA with 5% loading yields was slightly higher than that of 10% treatment, wherein only 80, 74, and 78% were achieved, respectively. Based on these results, we have drawn an interpretation of optimized loading of acid for the treatment as 5%. This chemical treatment improved not only the oil quality in physical appearance but also the characteristic properties, which are discussed in consecutive sections.

The chemical upgrading of WPO using acid treatment proceeds primarily through acid-catalyzed reactions involving unsaturated and aromatic hydrocarbons. Strong acids, such as sulfuric acid (SA) and methane sulfonic acid (MSA), protonate olefinic bonds, generating carbocation intermediates that undergo oligomerization, polymerization, and condensation reactions. These reactions convert reactive low-molecular-weight species into heavier, insoluble fractions that separate as sludge. Additionally, the acids facilitate the removal of oxygenated compounds and moisture through dehydration reactions, thereby improving the hydrocarbon purity and calorific value of the oil. The effectiveness of the treatment depends on the acid strength, with SA and MSA exhibiting higher activity compared to HCl and acetic acid (AA), resulting in greater reductions in viscosity, density, and impurity content.

Residual acidity is an important consideration for chemically upgraded pyrolysis oils because acidic compounds may affect storage stability, corrosion behavior, and long-term engine durability. In the present study, TEA neutralization was performed following acid treatment to minimize residual acidic species. Although the neutralization and purification steps significantly reduced acidity, trace acidic compounds may still remain in the upgraded oil. Therefore, detailed acid number evaluation and long-term corrosion studies are necessary for future large-scale implementation.

3.2. Treatment Effect on Density and Viscosity

The variation in density and viscosity of the upgraded oil has also been studied to draw an interpretation for the most effective chemical treatment agent. Maximum reduction in the density was achieved (0.789 g/cc) for SA treatment on WPO,

Table 2. Specifications of Diesel Engine Used for Conducting Engine Performance and Combustion Characteristics[24]

make and model	Kirloskar TV1
Engine type	variable compression ratio (VCR) engine
No. of cylinder	single cylinder
type	four-stroke
type of cooling	water-cooled
fuel type	diesel fuel
bore (mm)	87.5
stroke (mm)	110
engine speed (rpm)	1500
rated power	7 bhp, 5.2 kW at 1500 rpm
compression ratio	16.2:1
static injection timing	23 deg before TDC (top dead centre)
loading arrangement	Eddy current dynamometer
type of fuel injection	direct injection

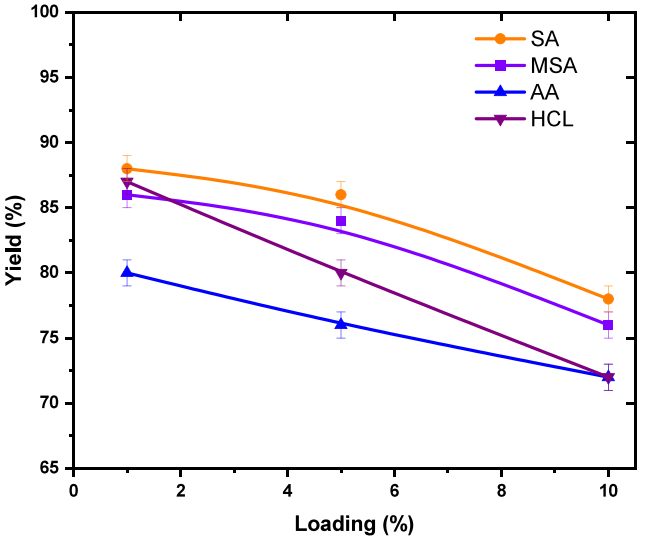


Figure 2. Yields of UWPO with different treating agents.

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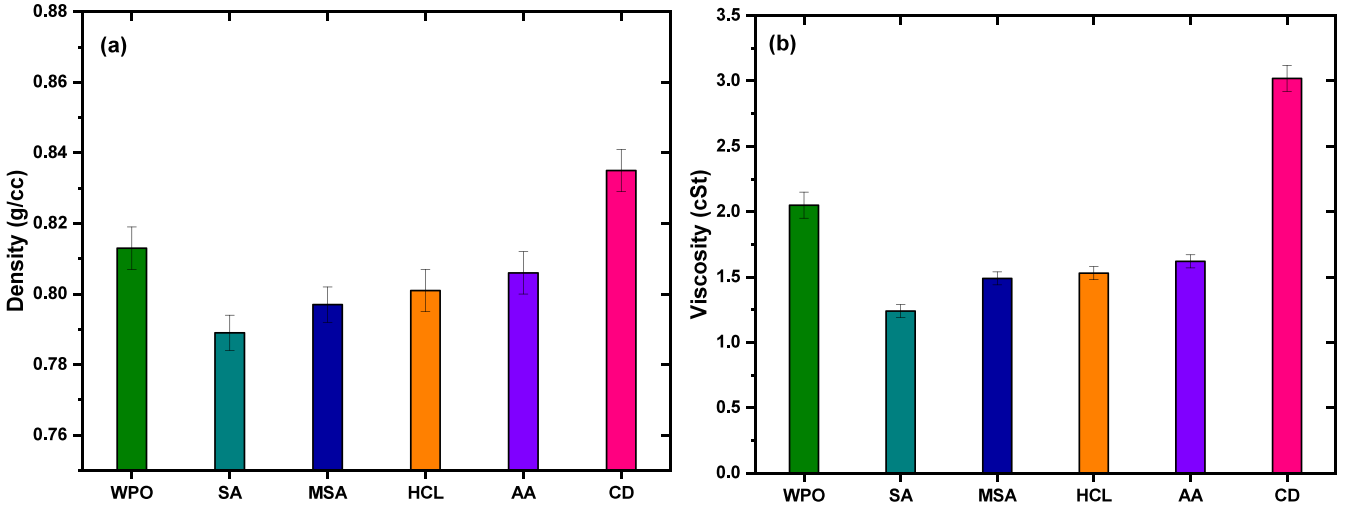


Figure 3. (a) Density and (b) viscosity of WPO and UWPO with different treating agents.

followed by MSA. The two acids have no significant change in density. This effect on density is shown in Figure 3 for an optimized chemical treatment of 5% on WPO, followed by base treatment. The change in viscosity also followed a similar trend to that of density, wherein the initial viscosity of WPO was found to be 2.05 cSt, which further reduced by 39.5% with SA treatment, while the other treatments on WPO resulted in a reduction of 27.3, 25.3, and 20.9% using MSA, HCl, and AA treatments, respectively. The attained density and viscosity by SA were similar to that reported for oil obtained through pyrolysis of PP waste by Sivagami et al. The UWPO has a lower density and viscosity compared to that of diesel.[25]

3.3. Treatment Effect in CHNSO

The ultimate analysis of the WPO and UWPO was performed, and the results are tabulated in Table 3. It revealed that the hydrocarbon carbon content of the oil increased, while the oxygen decreased in general for all the chemical treatment processes. The SA treatment provided a higher carbon content (86.96%) as compared to the other acid treatments and WPO. The achieved carbon content in the UWPO is lower than that of commercial diesel.

3.4. Treatment Effect in Chemical Composition

The detailed hydrocarbon analysis PIONA for the WPO and UWPO obtained is shown in Figure 4. The composition of the treated oil, the paraffin, naphtha, and olefin content increased in the treated oil, while the aromatics and isoparaffins reduced when compared with raw oil, and it is observed that the composition of the oil treated with SA has a lower paraffin content (16.78%) and a higher olefin content (31.36%) than that of oil treated with MSA (21 and 29%) respectively. Similarly, the gasoline (C_5–C_{12}) content increased in the treated oil, while the other composition of lighter diesel (C_{13}–C_{18}) and heavy diesel (C_{19}–C_{25}) fraction reduced.

Table 3. Ultimate Analysis of Pyrolysis Oil and Treated Oil (wt %)

		UWPO				
element	WPO	SA	MSA	AA	HCL	diesel
C	82.26	86.96	84.97	85.11	84.47	90.06
H	11.49	12.71	14.49	14.27	14.21	9.72
N	0.257	0.09	0.27	0.68	1.01	0.0681
S	0.080	0.05		0.043	0.051	0.0398
O	5.89	0.19	0.27	1.87	1.79	0.2560

3.5. Treatment Effect on Calorific Value

The calorific value of oil represents the energy released upon combustion and is a critical parameter in evaluating fuel quality. The calorific value of the oil was enhanced by the treatment, and it seems to be higher in the SA treatment compared to the other treatments. This enhancement can be attributed to the ability of SA to remove impurities such as moisture, free fatty acids, and non-combustible components, which dilute the energy content of untreated oil. Acting as a dehydrating agent, SA reduces moisture levels and concentrates energy-dense compounds.[26,27] Therefore, SA treatment proves to be an effective approach for enhancing the energy potential of oil from 35.57 to 44.75 MJ/kg. The physicochemical properties of WPO and UWPO are reported and tabulated in Table 4. This result shows that the upgraded oil has better physicochemical properties, particularly with respect to density, viscosity, and carbon residue, while the lower flash and fire point was obtained compared to diesel.[28]

The flash point and fire point values of UWPO were lower than those of conventional diesel fuel, primarily due to the presence of residual low-boiling and volatile hydrocarbon fractions generated during the pyrolysis process. Although the chemical upgrading treatment improved several fuel properties, lighter hydrocarbon components were not completely removed during the upgrading process. The presence of these volatile compounds reduced the temperature required for vapor ignition, thereby lowering both flash point and fire point values. Such behavior may raise safety concerns during storage and handling of neat UWPO. However, blending UWPO with diesel improves the overall safety characteristics of the fuel mixture.

3.6. Engine Performance Analysis

The engine performance of the fuels depended on the physicochemical properties of the fuels. However, the physicochemical

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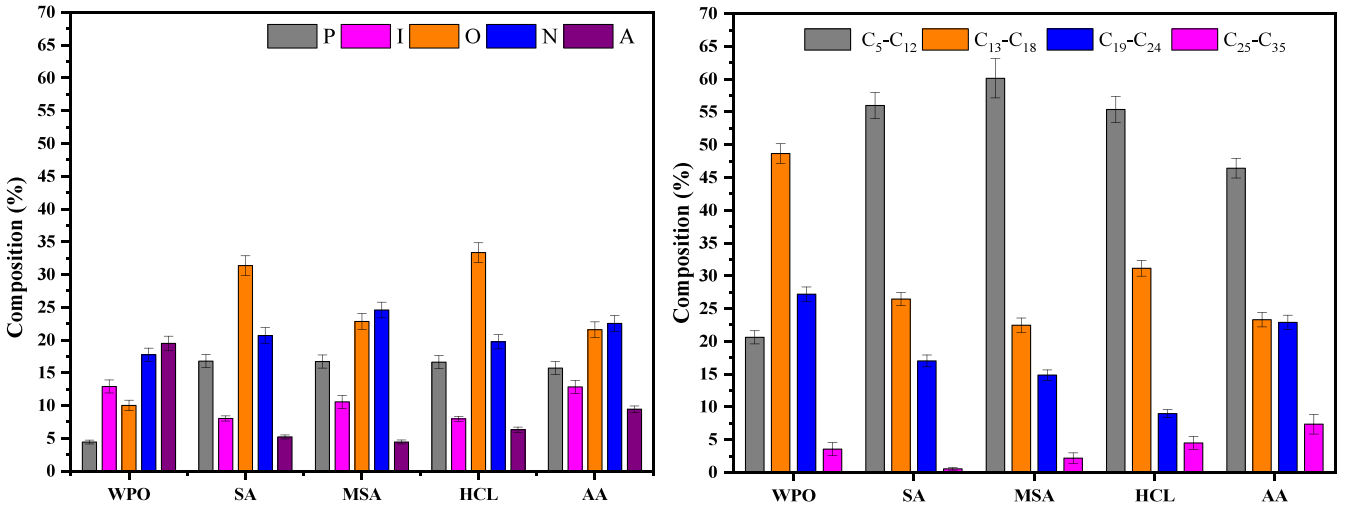


Figure 4. Semi-quantitative hydrocarbon group distribution (PIONA-type analysis) obtained from GC-MS of WPO and UWPO treated oil.

Table 4. Physicochemical Properties of UWPO Compared to WPO and CD

		UWPO		
specification	WPO	SA	MSA	diesel
density (g/cc)	0.8931	0.7844	0.7962	0.8350
viscosity cSt (K)	1.89	1.08	1.81	3.02
pour point (°C)	3	−4	−4	−20
flash point (°C)	10	10	12	80
fire point (°C)	15	14	15	82
HHV (MJ/kg)	35.57	44.75	42.41	42.04
water content (ppm)	81.0	62.0	148.0	200.0
carbon residue (%)	0.10	0.01	0.10	0.18
chlorine content (%)	0.90	0.43	0.90	0.06
ash content (%)	0.092	0.01	0.10	0.01

properties of blended fuels may vary with the different blending ratio.[24,29] Moreover, the heating values and fluidity properties of the fuels influence the brake thermal efficiency of the machine. The change in brake thermal efficiency (BTE) with varying engine loadings of CD100, UWPO10, and UWPO20 are shown in Figure 5a. The BTE of all the fuel samples increases by increasing the loading conditions. However, the BTE for UWPO10 and UWPO20 samples was found to be higher up to 50% loading conditions due to a change in cylindrical pressure, resulting from the variation in physicochemical properties of the fuel, such as viscosity, calorific value, density, etc., of the blends.[30,31] Moreover, the BTE of UWPO10 was observed to be better than that of UWPO20 under full load conditions, which basically occurs due to a higher rate of pressure rise. The highest and lowest BTE were found to be 33.86 and 27.80% for CD and UWPO20, respectively. The BTE was found to be higher at intermediate loadings; however, a significant reduction (~7.2 and ~17.9% for UWPO10 and UWPO20, respectively) was seen in BTE under full load conditions. The lower viscosity and higher calorific values of the fuels subsequently improve the BTE.[31−34]

The variation in fuel consumption with varying loadings for all tested fuels (CD100, UWPO10, and UWPO20) is shown in Figure 5b. It is majorly impacted by the fuel characteristics (viscosity, density, and API gravity) during fuel combustion.[30,31,33] The engine experiments revealed that BSFC is decreased when the load was increased from 0–100%. From the current investigations, BSFC was found to be the lowest (6.353×10^{-5} kg/kW s) for CD100 and highest (7.804×10^{-5} kg/kW s) for the UWPO20 fuel sample under 100% load conditions. The BSFC of the fuels decreased from 1.872×10^{-4} to 6.353×10^{-5} kg/kW s for CD, from 1.7×10^{-4} to 6.869×10^{-5} kg/kW s for UWPO10, and from 1.773×10^{-4} to 7.804×10^{-5} kg/kW s for UWPO20, respectively. The high fuel consumption for UWPO blended fuels was observed due to the high viscosity of UWPO oil, as compared to CD100.[27,35,36] Comparing the BSFC of UWPO blended fuels, it was found to be 7.51 and 18.6% higher than the CD100 under 100% load conditions. Moreover, the percentage reduction in BSFC was found to be 66.06, 59.6, and 55.97%, respectively, for CD100, UWPO10, and UWPO20 when the engine load was increased from 25 to 100%.

The engine exhaust gas temperature with varying load conditions for all tested fuels is presented in Figure 5c. The results showed that EGT was increased with the loading conditions for all tested fuels. The variation of EGT was found to be in the range of 171.45–249.43, 154.45–247.33, and 156.93–247.08 °C, respectively, for CD100, UWPO10, and UWPO20 fuel samples. However, the EGT profiles showed the highest values for CD100 under all loading conditions, which is attributed to the higher in-cylinder pressure during the combustion process.[29] Figure 5d shows the variation in brake power (BP) under various loading conditions for CD100, UWPO10, and UWPO20. The results revealed that BP increases with engine loading conditions irrespective of the type of fuel blending ratios and fuel properties. The estimated BP was found to be in the range of 0.04–3.28, 0.03–3.35, and 0.06–3.33 kW for CD100, UWPO10, and UWPO20, respectively, when the load was varied from 0 to 100%. The analysis showed a similar trend for all blended fuel samples, and it is comparable with CD100.

Mechanical efficiency (ME) is one of the influencing parameters during engine testing of fuels. It is the ratio of brake power and indicated power as expresses.[24,37] The variation of ME with

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varying load conditions for all tested fuels (CD100, UWPO10, and UWPO20) is shown in Figure 5e. It showed that the ME increased by increasing the engine load. However, the mechanical efficiencies with UWPO10 and UWPO20 fuels were found to be higher compared to CD100 under full load conditions. The maximum recorded ME values at full load conditions were found to be 87.47, 92.87, and 88.59%, respectively, for CD100, UWPO10, and UWPO20. Blending of UWPO increases the ME by 5.81 and 1.29%, respectively, for UWPO10 and UWPO20 under full load conditions.

The air–fuel (A/F) ratio is one of the important characteristics of engine performance analysis, which helps us to identify the nature of the combustion of the fuel.[37] It is the ratio of air intake and fuel flow rate during the engine operation. The

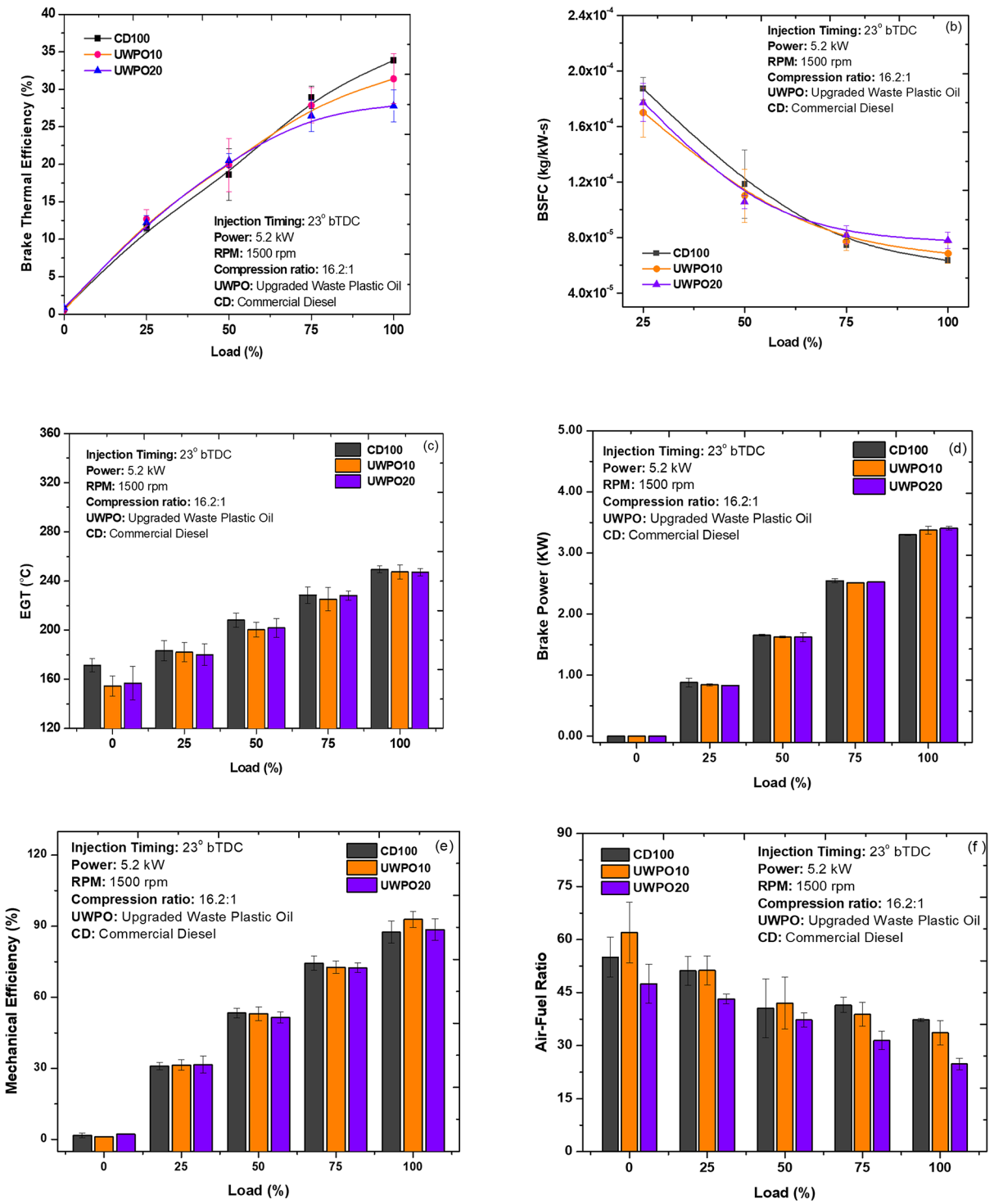


Figure 5. Engine performance characteristics with varying load conditions: (a) brake-specific fuel consumption, (b) brake thermal efficiency, (c) exhaust gas temperature, (d) brake power, (e) mechanical efficiency, and (f) air–fuel (A–F) ratio.

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influence of A/F ratio at various loading conditions has been investigated, and the results are depicted in Figure 5f. The results showed that the A/F ratio decreases with increased loading conditions due to change in the fuel flow rate. The A/F ratio differs between 55.02–37.25, 62.02–33.63, and 44.27–24.78 for CD100, UWPO10, and UWPO20, respectively. The reduction in A/F ratio was found to be 32.29, 45.79, and 47.83% for CD100, UWPO10, and UWPO20 respectively, when load changes from 0 to 100%. The lower A/F ratio and higher A/F % reduction indicate the higher emissions of CO, CO_2, and UHC pollutants during fuel combustion, and it is attributed to higher fuel consumption.[37,38] The Engine Combustion Analysis (Section S1) and Engine Exhaust Gas Emission Analysis (Section S2) for the oil blends have been performed and have been reported in the Supplementary Information.

3.7. Comparison with Recent Studies

The physicochemical properties of the upgraded WPO (UWPO) obtained in this study are comparable with those reported in recent literature on upgraded plastic pyrolysis oils. The observed reduction in viscosity and density, along with an increase in calorific value (~44.75 MJ/kg), falls within the range reported for catalytically upgraded and hydrotreated pyrolysis oils, although such methods typically involve significantly higher temperatures and pressures.[18,39,40] In contrast, the present study achieves comparable improvements under mild operating conditions, highlighting the effectiveness of the proposed chemical upgrading strategy.

In terms of engine performance, the decrease in brake thermal efficiency (BTE) and increase in brake specific fuel consumption (BSFC) at higher blend ratios (UWPO20) are consistent with previously reported trends for plastic pyrolysis oil diesel blends.[41−43] These deviations from diesel performance are generally attributed to differences in volatility, atomization behavior, and fuel composition. Notably, the relatively improved mechanical efficiency observed for the UWPO10 blend aligns with findings reported in earlier studies, where moderate blending ratios exhibited more stable combustion characteristics and better engine adaptability.[42,43]

The emission characteristics observed in this study also follow trends reported in the literature. The increase in CO and HC emissions for UWPO blends can be attributed to incomplete combustion and the presence of residual unsaturated hydrocarbons, which has been widely reported in previous studies on pyrolysis oil utilization in diesel engines.[41−43] Similarly, the observed increase in NOx emissions is consistent with findings linking higher in-cylinder temperatures and altered combustion dynamics to NOx formation.[41,43]

The sulfur content of waste plastic pyrolysis oil is largely dependent on the nature of the feedstock and the presence of sulfur-containing additives, pigments, or contaminants in the plastic waste stream. In the present study, the acid treatment was employed primarily for impurity removal and did not involve sulfur-containing reagents; therefore, no direct sulfur incorporation into the upgraded oil is anticipated. However, the removal of suspended carbon, ash-forming constituents, and other non-sulfur impurities may alter the relative concentration of sulfur-containing compounds in the liquid phase. Consequently, any observed variation in sulfur content after treatment is more likely associated with compositional redistribution rather than sulfur addition. Further detailed sulfur speciation studies using advanced analytical techniques would provide deeper insights into the behavior of sulfur-containing compounds during the upgrading process.

From an upgrading perspective, conventional approaches such as hydrotreatment and catalytic upgrading have demonstrated significant improvements in fuel quality but require hydrogen input, elevated pressures, and complex reactor systems.[40,42] In contrast, recent studies have emphasized the need for simpler and more energy-efficient upgrading strategies.[16,18] The present work contributes to this direction by demonstrating that a low-temperature, acid-mediated chemical upgrading approach can yield fuel properties and engine performance comparable to more intensive processes.

Overall, the results indicate that the proposed method provides competitive performance relative to existing upgrading strategies while offering advantages in terms of operational simplicity and reduced energy requirements. These findings support the potential of chemically upgraded WPO as a viable blending component for diesel fuel within a sustainable waste-to-energy framework.[16,18,39−43]

The present study, while highlighting the potential of upgraded waste plastic pyrolysis oil (UWPO) as a sustainable alternative fuel, has certain limitations. The analysis was restricted to lower blend ratios (10 and 20%) with diesel, and the performance of higher blends or pure UWPO remains unexplored primarily to ensure compatibility with engine operation and to avoid drastic deviations in fuel properties that could negatively affect combustion stability, efficiency, or safety. The results are engine-specific and may vary across different engine types and operating conditions, limiting generalization. Long-term impacts on engine durability, injector fouling, and lubrication compatibility were not assessed, leaving uncertainties regarding extended use. Additionally, the work focused mainly on performance and basic emission trends, without a comprehensive analysis of regulated and unregulated pollutants, such as NOx, SOx, particulate matter, or unburned hydrocarbons. Storage stability and potential degradation of upgraded oil under prolonged conditions were also not considered. Furthermore, the study was conducted at a laboratory scale, without addressing large-scale process economics, logistics, or the variability of feedstock quality that can influence pyrolysis oil properties. These limitations suggest the need for more extensive research on engine compatibility, emission profiling, storage behavior, and techno-economic feasibility to advance UWPO as a viable circular fuel solution.

Although the chemical upgradation process substantially improved the quality of waste plastic pyrolysis oil by reducing suspended carbon and other impurities, trace amounts of residual contaminants and heavier hydrocarbon fractions may remain in the upgraded fuel. These species have the potential to contribute to injector fouling, combustion chamber deposits, fuel filter blockage, and lubricant deterioration during prolonged engine operation. The present study was limited to short-term performance and emission evaluation; therefore, long-term durability aspects such as injector cleanliness, wear of fuel system components, deposit accumulation, and engine reliability were not investigated. Future studies should focus on extended engine endurance testing, deposit characterization, lubricant analysis, and component wear assessment to establish the long-term suitability of upgraded waste plastic pyrolysis oil blends for commercial engine applications.

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3.8. Techno-Economic Assessment

The techno-economic assessment (TEA) for production of upgraded oil is estimated considering the following proposed schematic flow diagram of the 10-TPD process plant, as shown in Figure 6.

3.8.1. Total Capital Investment (TCI). The total capital investment (TCI) for the plant comprises the fixed capital investment (FCI) and the working capital investment (WCI). The FCI is divided into two categories: direct and indirect costs. Direct costs include purchased equipment, installation, instrumentation and controls, piping, buildings, yard improvements, service facilities, land, and electrical systems. Indirect costs cover know-how and technology fees, engineering and supervision, construction expenses, contractor fees, legal charges, contingencies, and start-up costs. Since many of these components are difficult to estimate precisely, their values were chosen from established ranges reported in the literature (Munagala et. al.,[19] Razak et al.,[20] and Munagala et al.[21]) and normalized for the present case. Based on equipment cost data provided by an engineering company for setting up a full-scale process plant with a maximum capacity of 11.63 TPD of raw material, the total equipment cost was estimated at ₹14,000,000. The WCI is typically considered as a fraction of FCI, and in this study, it was taken as 17.6% of FCI. Accordingly, the total capital investment (TCI) for a 10 TPD Oil Upgradation plant with an assumed oil yield of 86% was estimated to be ₹47270588.24, as detailed in Table 5.

3.8.2. Total Production Cost (TPC). The total production cost comprises both manufacturing costs and general expenses. Manufacturing costs are further divided into variable production costs, fixed charges, and plant overheads. Variable costs include raw material, operating labor, utilities, office and clerical expenses, maintenance and repairs, operating supplies, and laboratory charges, while the feed WPO was taken as ₹40. Fixed charges account for depreciation, local taxes, insurance, and plant overhead, while general expenses cover administrative overhead, marketing/distribution and selling costs, and research and development activities. For the present case, the process plant operates 330 days per year, producing approximately 3300.6 ton (4,209,941 L) of oil. A detailed breakdown of the annual production cost is presented in Table 6. Under the given conditions, the total production cost was estimated at ₹197,744,035. Based on market specifications, the assumed selling prices of the products are ~₹65 L^{-1} for oil.

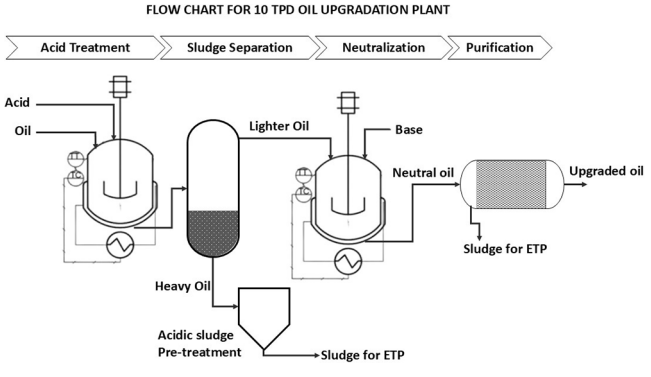


Figure 6. Schematic diagram used for techno-economic evaluation of 10-TPD pyrolysis oil upgradation to fuel plant.

Table 5. Components of Total Capital Investment (TCI) and Their Estimated Cost for the Oil Upgradation Plant for a Capacity of 10 TPD

total capital investment (TCI)	estimated cost (₹)
fixed capital investment (FCI)	
(A) direct cost	
purchased equipment	14,000,000
purchased equipment installation	1,866,133
instrumentation and controls	3,732,265
piping	3,111,556
electrical	2,174,485
building (including services)	3,111,556
yard improvements	156,178
service facilities	3,111,556
land	620,709
total direct cost	31,884,439
(B) indirect cost	
engineering and supervision	1,553,776
construction expenses and contractor fee	1,245,423
legal expense	156,178
contingency	1,553,776
start-up expenses	1,241,419
total indirect cost	5,750,572
total fixed capital investment (FCI)	40,180,000
working capital (WCI)	7090588.23
total capital investment (TCI)	47270588.24

3.8.3. Economic Analysis. The economic performance of the present process was evaluated using key indicators such as net profit, rate of return (ROR), and payback period for the overall yield of 86% for the fuel oil. Net profit was calculated as the difference between total annual sales and production costs, ROR as the ratio of net profit to total capital investment, and payback period as the ratio of fixed capital investment to

Table 6. Components of Total Production Cost (TPC) per Year for 10-TPD Pyrolysis Oil Upgradation to Fuel Plant

total production cost (TPC)	estimated cost (₹)
(A) manufacturing or operating cost	
(i) variable production cost	
raw materials	178,940,355
operating labor (plant manager-1, operator-2, skilled labor-2, unskilled labor-2)	3,360,000
utilities	2,520,000
maintenance and repairs	803,600
operating supplies	80,360
laboratory charges	336,000
(ii) fixed charges	
depreciation	4,099,200
local taxes	401,800
insurance	160,720
(iii) plant overhead	1,680,000
(B) general expenses	
administrative cost	504,000
distribution and selling cost	504,000
total product cost (TPC)	197,744,035

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annual profit. These parameters were estimated using the data from the capital investment and production cost analysis at different operating capacities, as summarized in Table 7, for a 10 TPD pyrolysis oil upgradation to fuel plant. The results indicate that at full operational capacity, the plant achieves a rate of return of nearly 120% with a payback period of 1 year 86% overall UWPO yield. At 10 TPD operation, the net annual profit from the combined sale of oil is estimated at ₹53,131,506 INR. Overall, the assessment confirms that the pyrolysis oil upgradation to fuel plant is economically viable and capable of generating value-added products with strong commercial potential.

4. CONCLUSIONS

This study thoroughly demonstrated the feasibility of employing chemically upgraded waste plastic pyrolysis oil (UWPO) as a diesel fuel additive. The upgrading process significantly enhanced critical fuel properties, yielding a density of 0.7844 g/cc, a gross calorific value of 44.75 MJ/kg, a fire point of 14 °C, a flash point of 10 °C, a pour point of −4 °C, and a viscosity of 1.08 cSt. These improved physicochemical properties provided the foundation for its evaluation in diesel engine applications. Engine performance analysis revealed that UWPO blends, particularly at 10% addition, achieved competitive brake thermal efficiency (92.87% mechanical efficiency) but displayed higher brake specific fuel consumption compared to commercial diesel, with values ranging from 6.353×10^{-5} kg/kW s for diesel to 7.804×10^{-5} kg/kW s for UWPO20. Combustion analysis further indicated measurable changes in cylinder pressure, rate of pressure rise, and heat release rate, signifying distinct combustion dynamics for the blends. Emission measurements showed increases in CO, CO_2, HC, NOx, and O_2 relative to neat diesel operation, reflecting the trade-offs associated with UWPO usage. Collectively, the findings highlight that chemically upgraded WPO, with improved fuel properties, has promise as an additive or partial substitute for diesel. However, the observed performance penalties and emission variations emphasize the need for further optimization, including detailed studies under variable compression ratios, long-term durability assessment, and emission mitigation strategies. The techno-economic assessment further confirmed process feasibility at scale. A 10TPD pyrolysis oil upgradation to fuel plant was estimated to generate a net annual profit of ₹5.32 Cr, with a rate of return of ∼120% and a payback period of 1–2 years. These results demonstrate that chemical upgradation of pyrolysis oil is both technically effective and economically attractive, paving the way for industrial-scale valorization of complex plastic waste into fuel-grade hydrocarbons for blend application. Addressing these aspects will be crucial to advance UWPO toward practical application, supporting sustainable energy development and contributing to effective plastic waste management.

Table 7. Parameters for Economic Analysis at 10 ton/day of Plastic Waste Processing with 330 Days Plant Operation

parameters	cost (₹)
manufacturing/operating costs	186,376,315
total production costs	197,744,035
profit before tax	75,902,151
net profit (30% tax)	53,131,506
rate of return (ROR)	120
payback period [Y]	1

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssusresmgt.6c00163.

Comprehensive additional details of the analyses like engine combustion and emission results (Section S1: engine combustion analysis and Section S2: engine exhaust gas emission analysis) and also supplementary figures (Figure S1: variation of combustion parameters with crank angle and Figure S2: emission characteristics of UWPO blended fuel samples and their comparison with commercial diesel at different loading conditions) (DOCX)

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Author Contributions

C.K.M.: experimental investigation, writing—original draft. B.A.: engine performance analysis, writing—original draft. S.M.R.: methodology, data curation. H.N.: validation, formal

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analysis. S.C.: resources, formal analysis. V.A.: supervision, writing—review and editing.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We thank the Director, CSIR-IICT (Ms. No. IICT/Pubs./2025/206), for providing all the required facilities to carry out the work, and the Department of Chemical Engineering, IISER Bhopal, for providing the fuel testing and engine characteristics analysis.

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ACS Sustain. Resour. Manag. 2026, 3, 2702−2713